Quaternary Logic Based Logic Circuits

Result Waveforms:

1.Standard Quaternary invertor

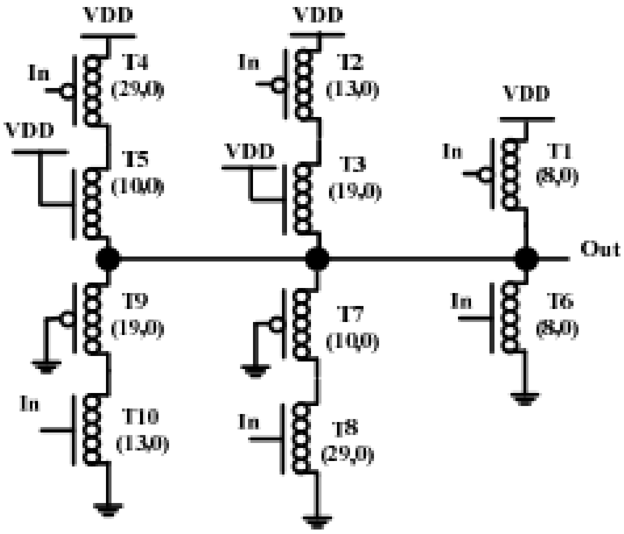
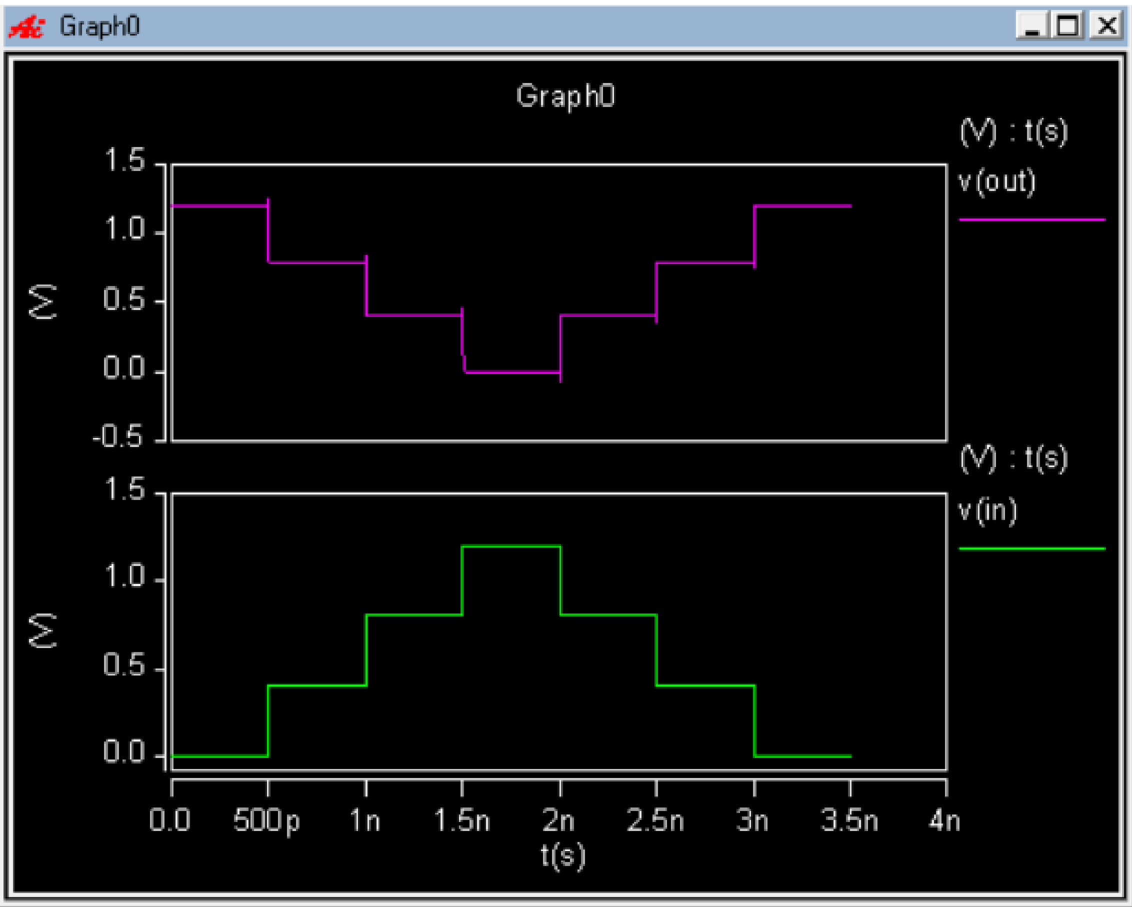


Fig. 6. Standard Quaternary Inverter [20]



2.Other Quaternary invertor

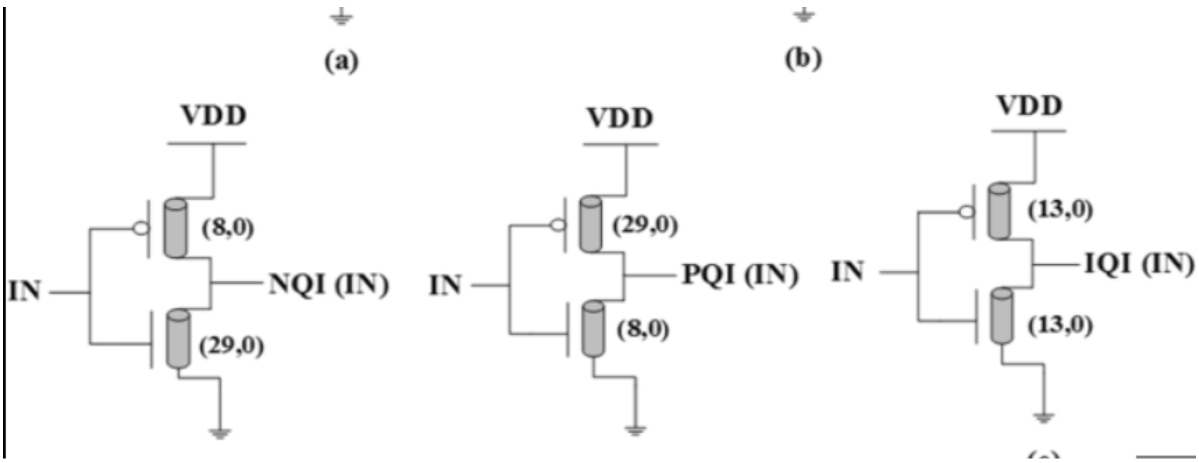
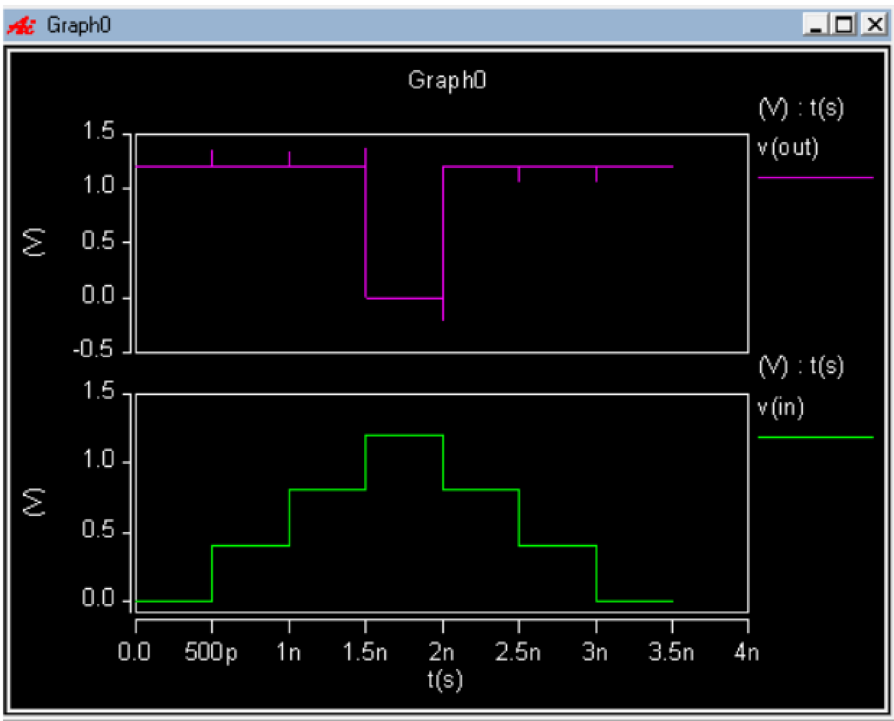


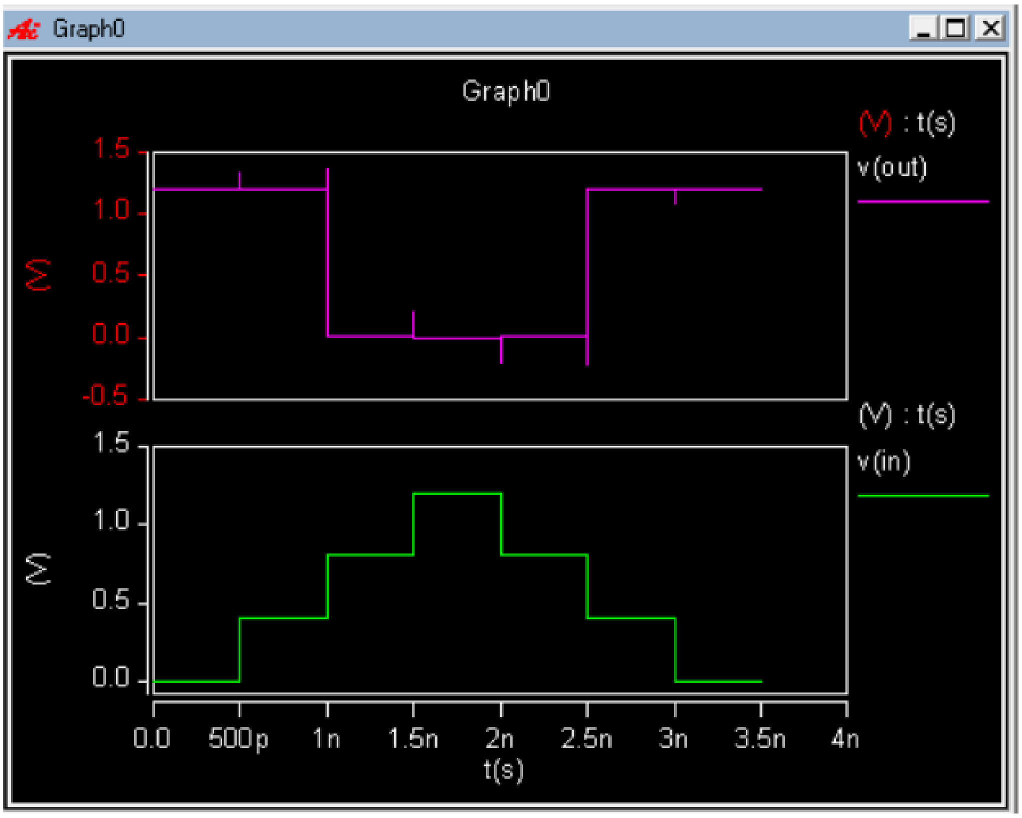
TABLE II. TRUTH TABLE OF QUATERNARY INVERTERS

Input	Inverter Output NOT(X)			
X	PQI	IQI	NQI	SQI
0	3	3	3	3
1	3	3	0	2
2	3	0	0	1
3	0	0	0	0

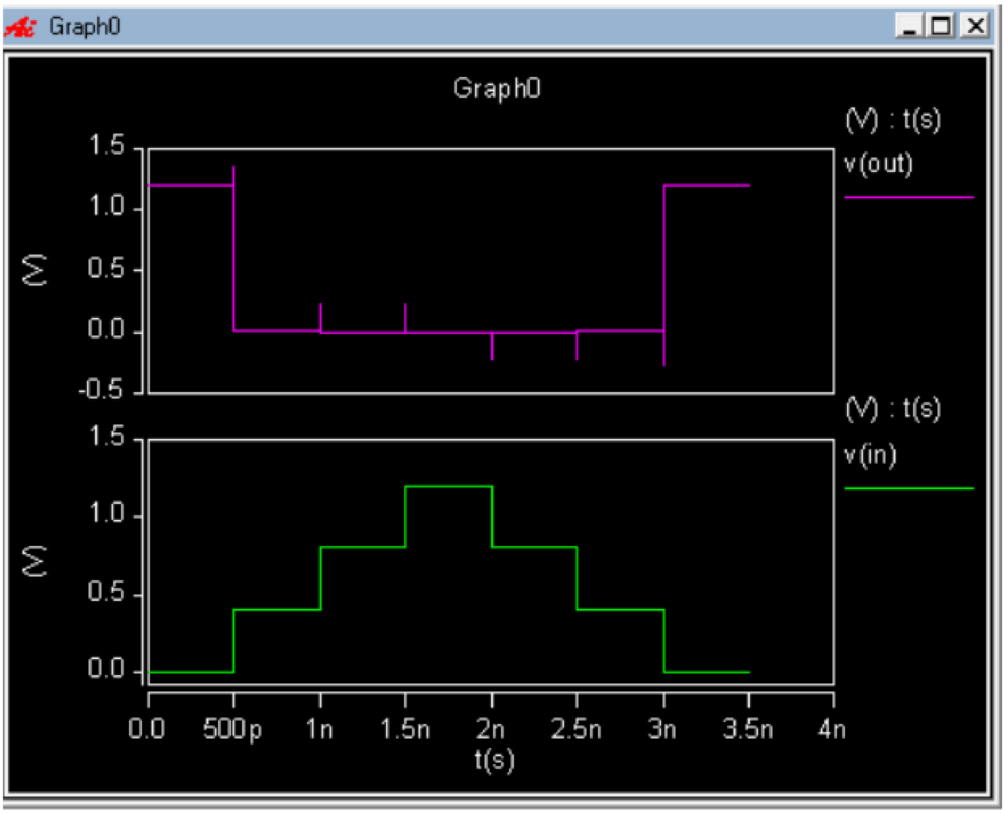
Positive Quaternary Invertor:



Intermediate Quaternary Invertor:



Negative Quaternary Invertor:



3.Quaternary QNand and QNor:

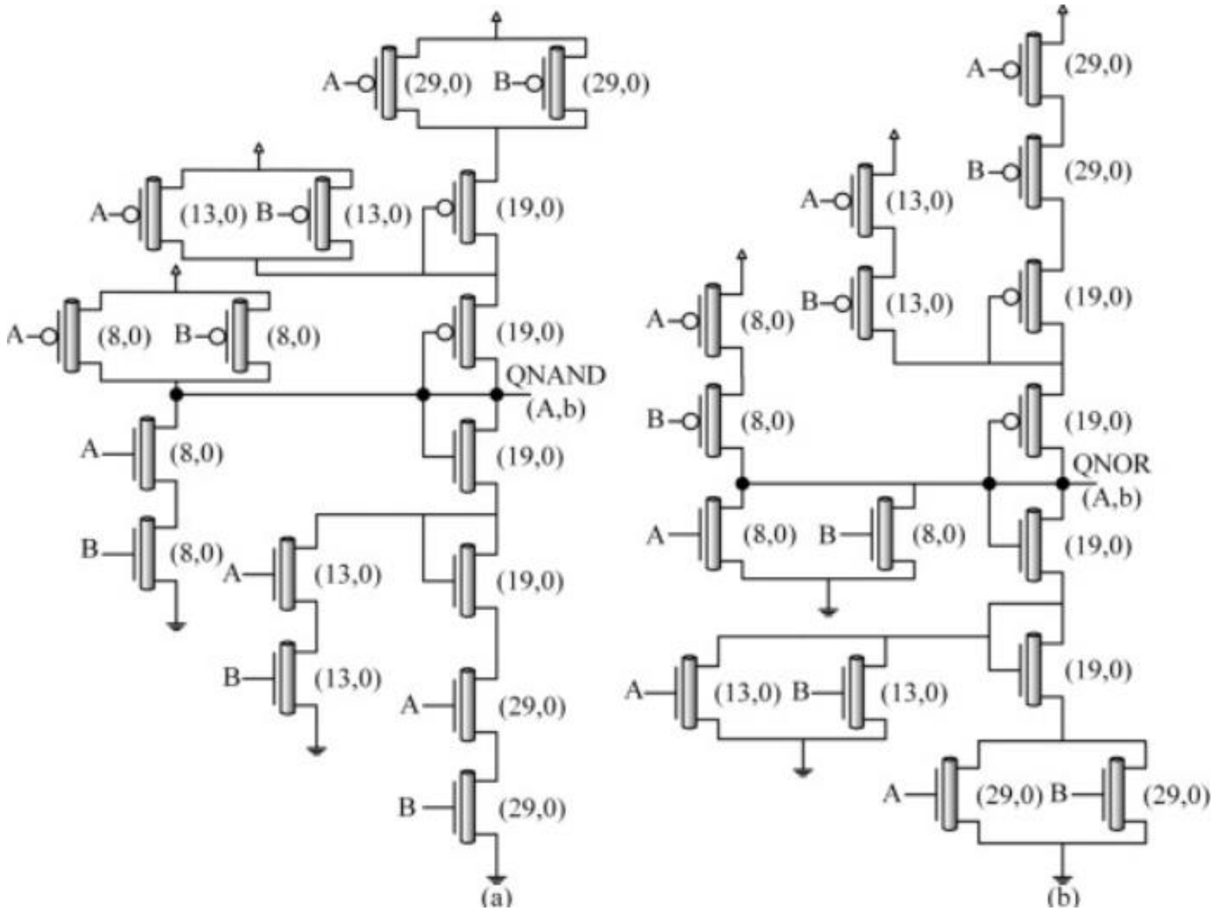
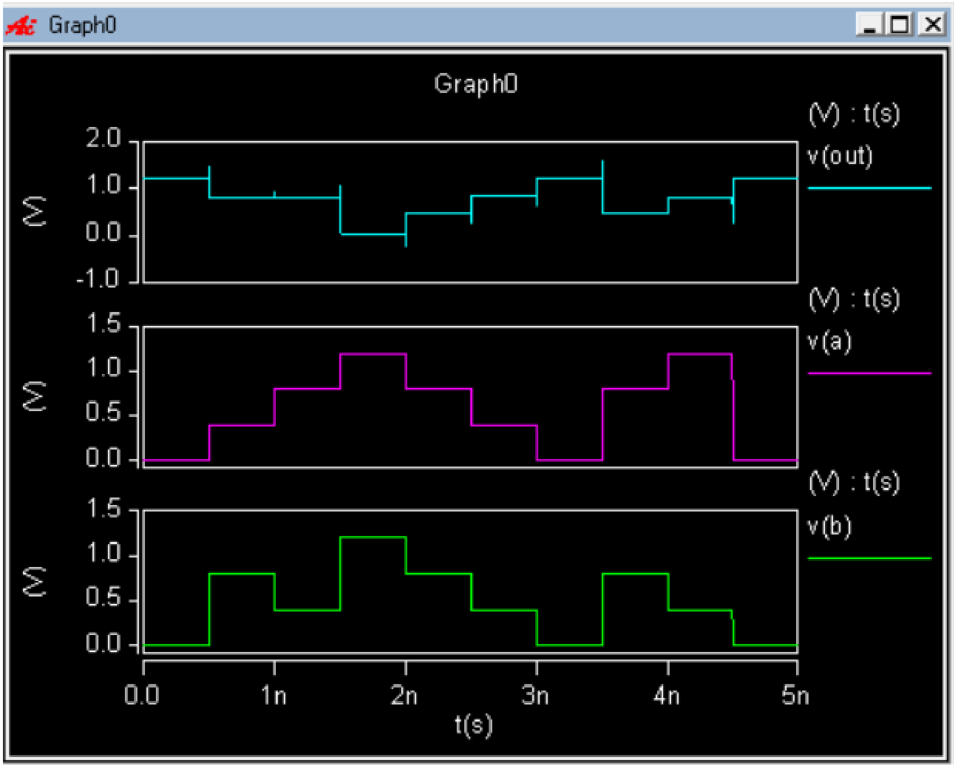


Table II: NAND and NOR Truth-tables

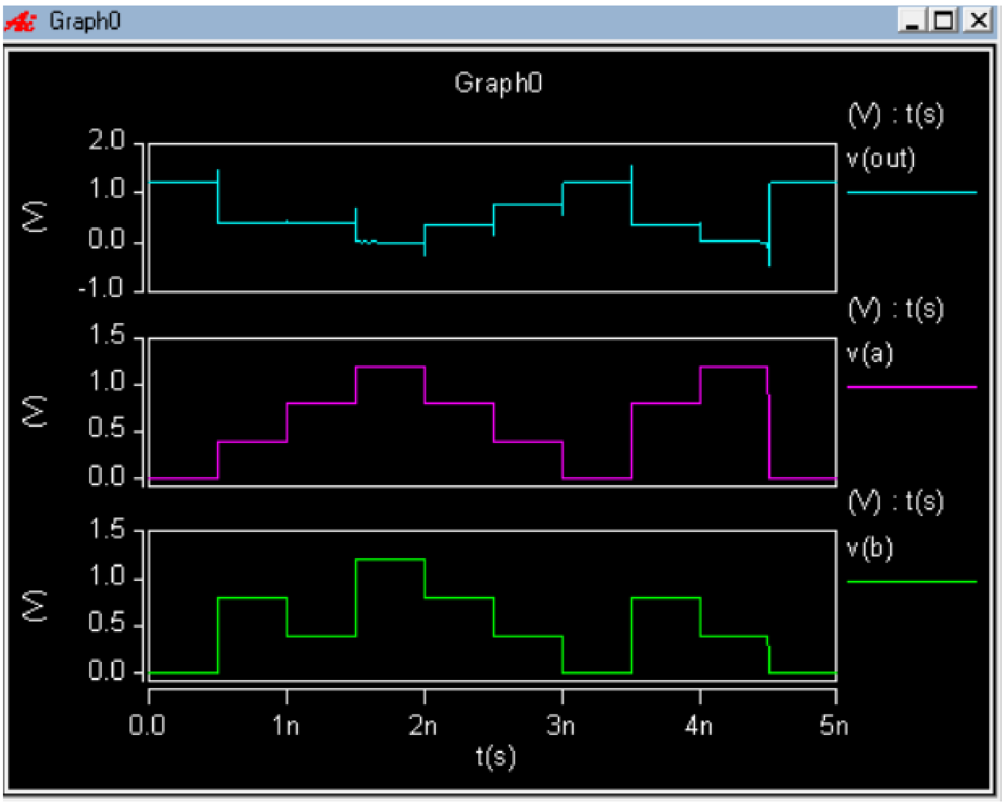
A	B	NAND	NOR
0	0	3	3
0	1	3	2
0	2	3	1
0	3	3	0
1	0	3	2
1	1	2	2
1	2	2	1
1	3	2	0

A	B	NAND	NOR
2	0	3	1
2	1	2	1
2	2	1	1
2	3	1	0
3	0	3	0
3	1	2	0
3	2	1	0
3	3	0	0

QNand:

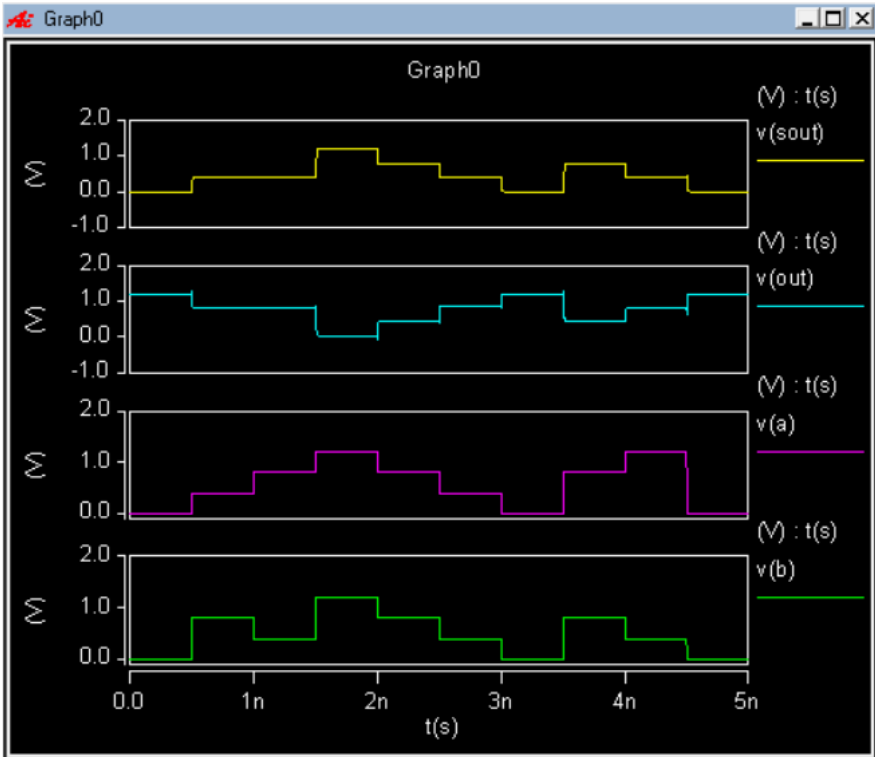


QNor:

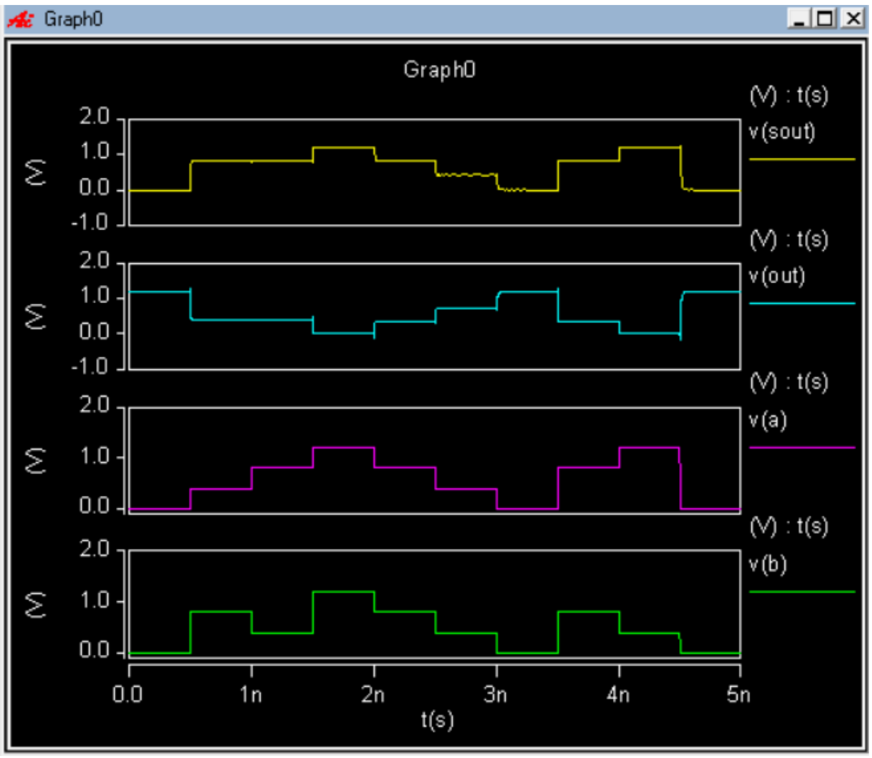


Quaternary Qand and Qor:

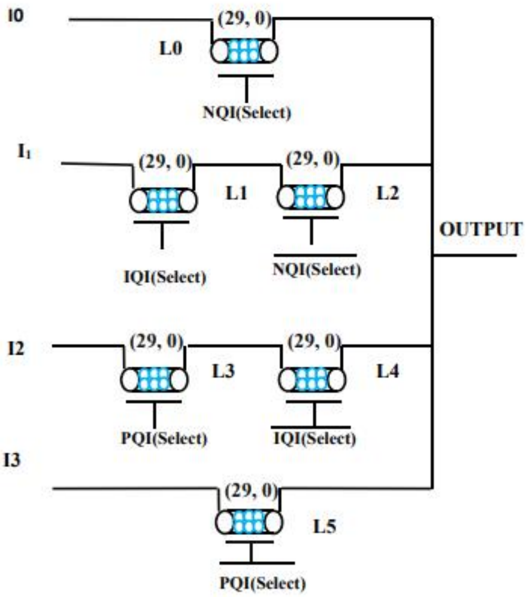
Qand:

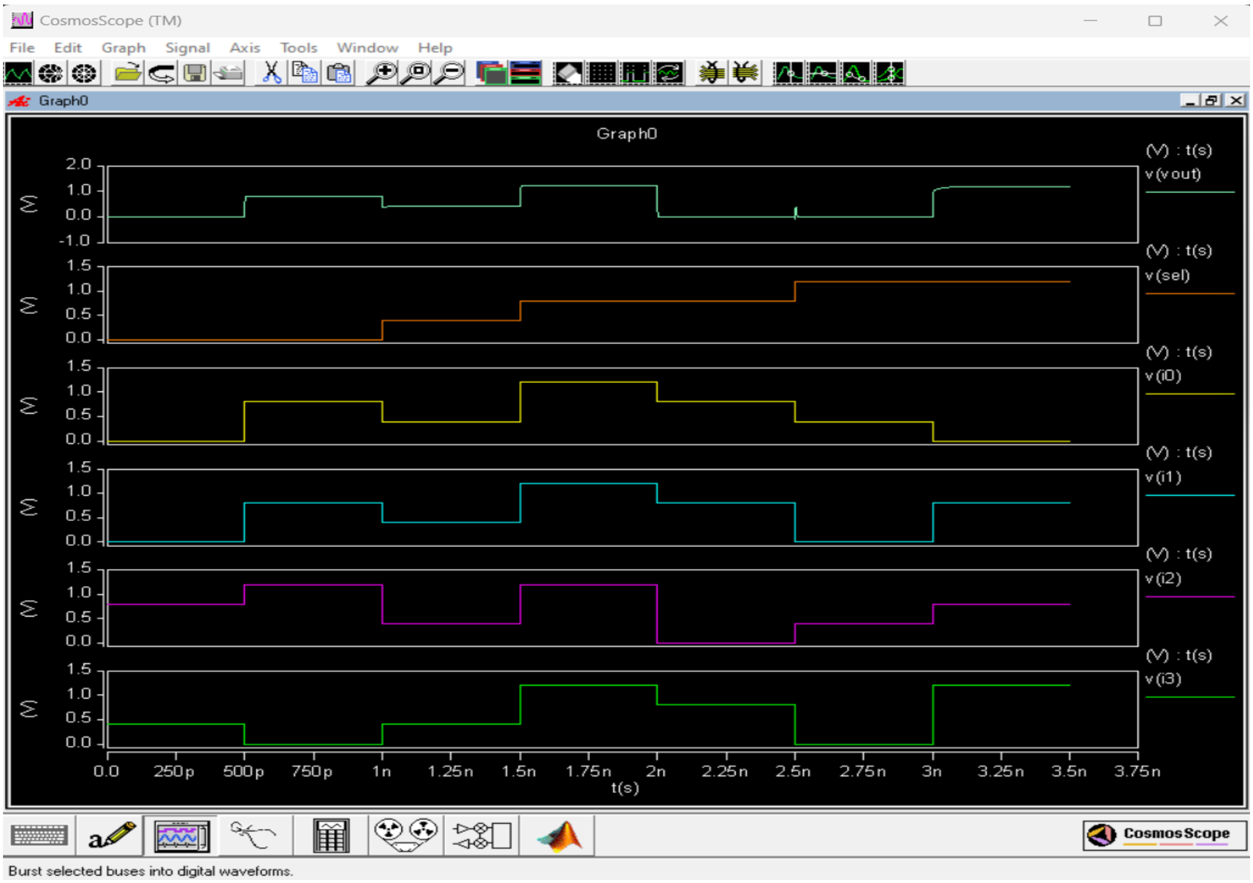


Qor:

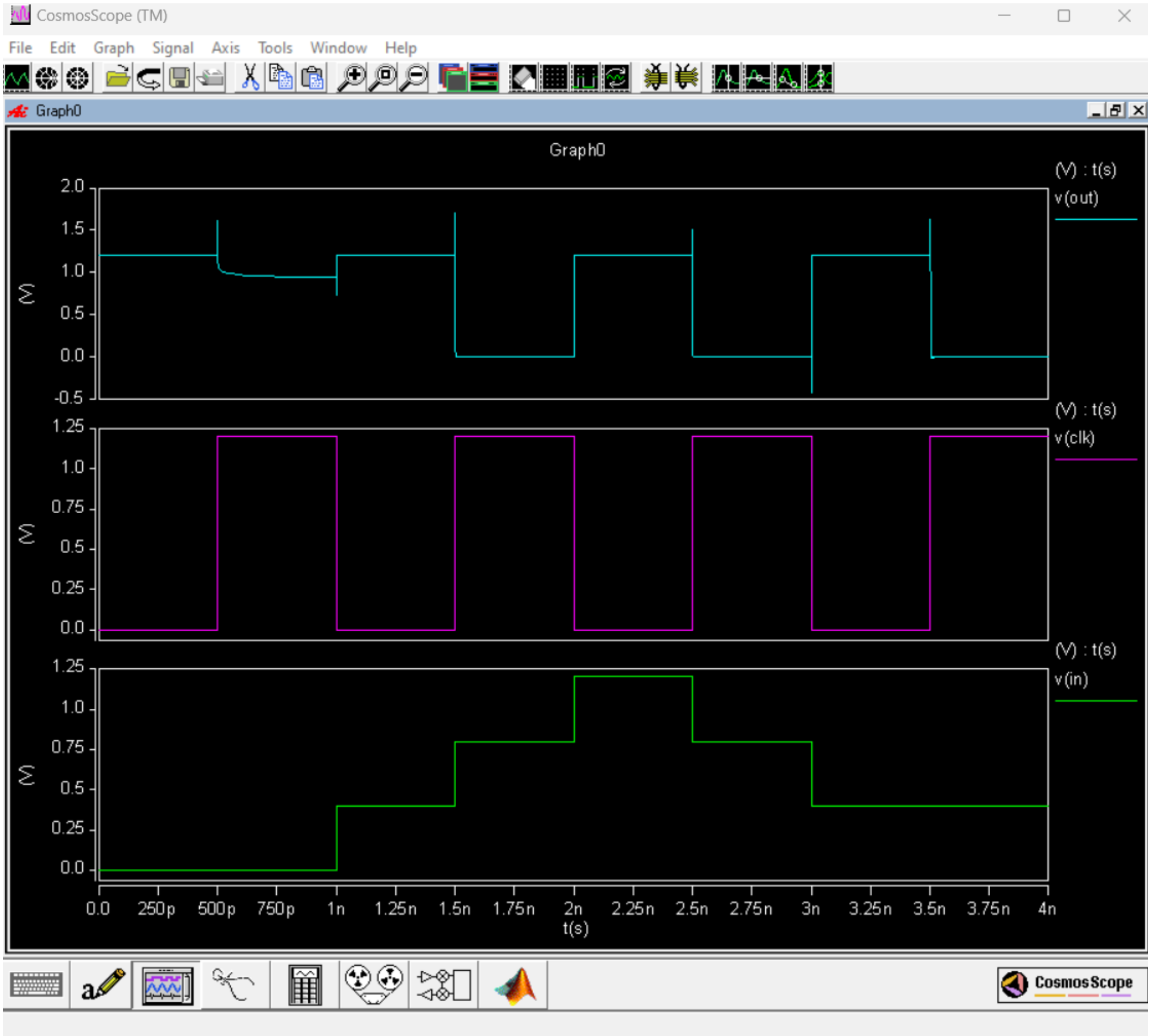


Multiplexer (4:1):

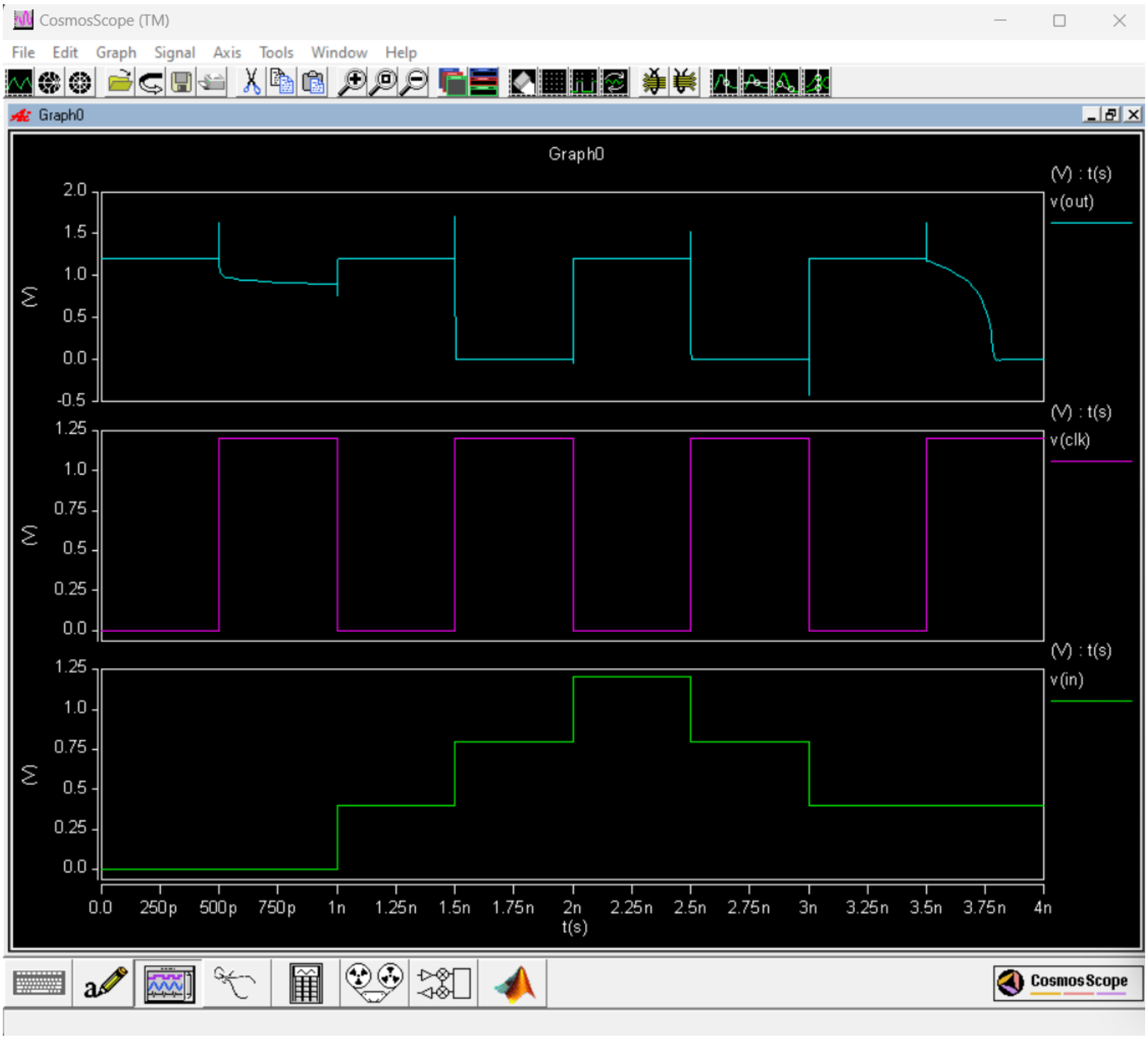




Dynamic Quaternary Negative and intermediate invertors:

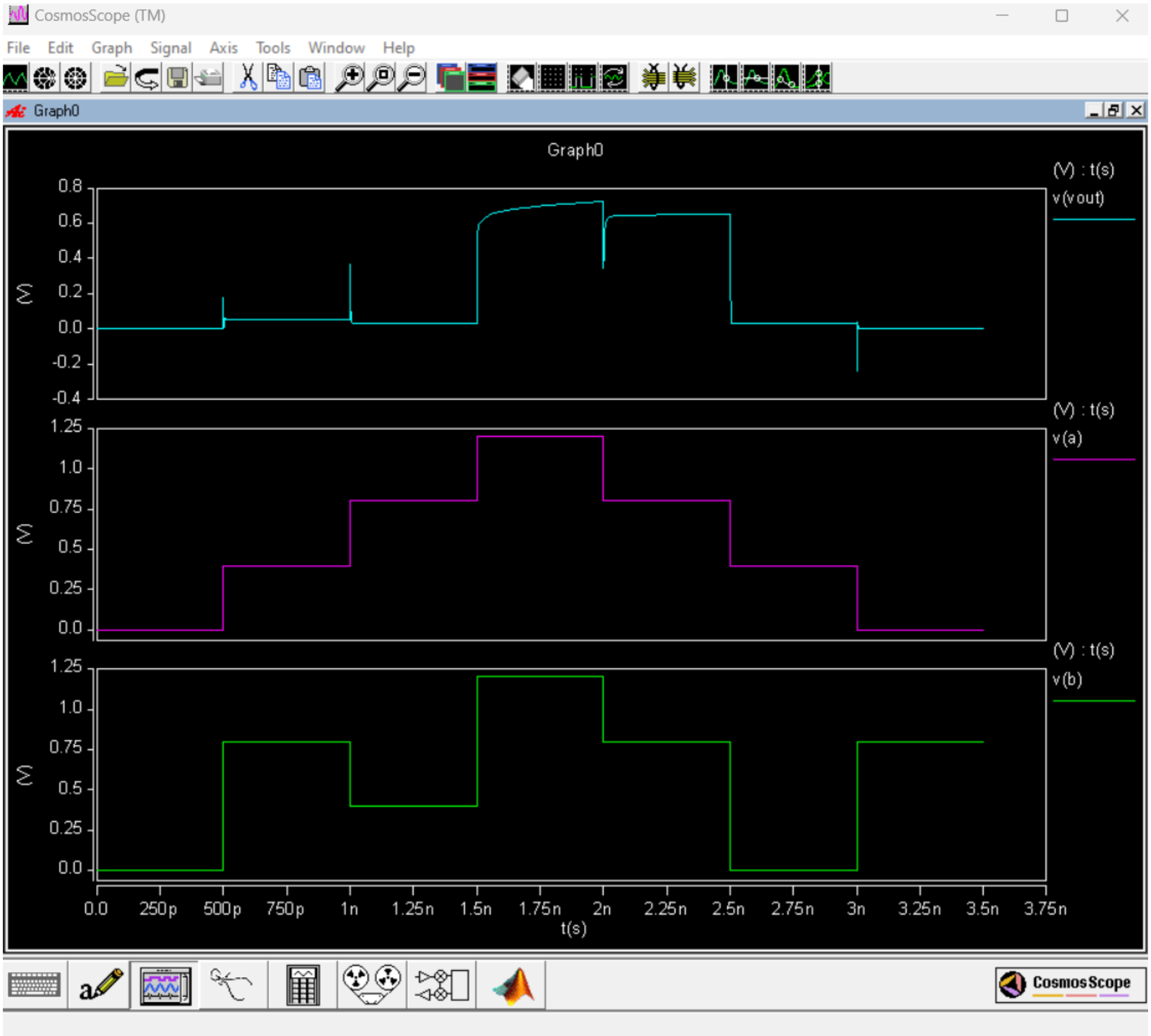

CosmosScope (TM)
File Edit Graph Signal Axis Tools Window Help
Graph0
Graph0
(V) : t(s)
v(out)
(V) : t(s)
v(clk)
(V) : t(s)
v(in)
t(s)
CosmosScope

Nqi dynamic



Iqi Dynamic

Half_adder Carry:



CosmosScope (TM)
File Edit Graph Signal Axis Tools Window Help
Graph0
Graph0
(V) : t(s)
v(vout)
(V) : t(s)
v(a)
(V) : t(s)
v(b)
(V)
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
1.25
1.0
0.75
0.5
0.25
0.0
0.0 250p 500p 750p 1n 1.25n 1.5n 1.75n 2n 2.25n 2.5n 2.75n 3n 3.25n 3.5n 3.75n
t(s)
CosmosScope